Continuous-Time Radar-Inertial Odometry with B-Spline Sliding-Window Optimization

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***Abstract*—Millimeter-wave radar offers an alternative when visibility limits optical sensing, but single-chip sensors provide sparse, noisy Doppler returns. This report investigates radar-inertial state estimation during indoor quadrotor flight using a single-chip 60 GHz radar (TI IWR6843) and a 1 kHz IMU, through flight regimes from hover to aggressive racing and aerobatic backflips at 10 rad/s. We first characterize the radar measurements against motion-capture ground truth. A per-return bearing-uncertainty law describes the Gaussian core, whose width grows with transverse speed and off-boresight angle. The errors also contain a frame-shared component and outliers. On the backflip flight, 38% of returns are Doppler-aliased. The estimator uses a continuous-time B-spline trajectory with one Doppler factor per return and a fixed-lag smoother with marginalization. A consensus front end rejects outliers, while the per-return law, frame covariance, and measured stream error budgets determine the weighting. The measurement-derived noise model and stream weights are held fixed across all evaluated flights and platforms. With reference yaw used as an idealized heading sensor, the causal live-edge velocity RMSE is 0.16–0.28 m/s and orientation RMSE is below 3° on racing, with position drift of 0.3–1.3% of path length. Backflip velocity and orientation RMSE rise to 1.66 m/s and 12.1°, reflecting both radar measurement limitations and the racing-oriented configuration. The solver runs in real time on a laptop CPU and reports conservative covariances on the racing flights. It transfers unchanged to a held-out fast-racing flight, but performance degrades on a separate held-out flight with extensive Doppler aliasing. On four public flights from a different platform with the same radar chip, velocity errors are comparable to EKF baselines without retuning the noise model or stream weights, although position drift is higher on three of the four flights. Radar and IMU with heading aiding thus provide a usable velocity and orientation estimate; absolute position drifts with distance.**

***Index Terms*—radar-inertial odometry, continuous-time estimation, B-spline, sliding window, marginalization, mmWave radar, Doppler velocity**

Declaration on the Use of AI Tools

Parts of the text of this report were formulated with the help of AI language models (Anthropic Claude, OpenAI ChatGPT, Google Gemini) from the author's notebooks, measurements, worklog and instructions, and were subsequently revised by the author throughout. Model wording is most concentrated in Section X (conclusion), Section VIII (results) and Section IV (system overview), at 30, 28 and 22 % of their lines. The models were also used as a discussion partner, for rewording and drafting text throughout, and for software development support during the project. All experiments and measurements are the author's own, and the author takes full responsibility for the entire content.

For transparency, a colour-coded copy of this report is published alongside it as `report/main_authorship.pdf` in the repository above, with a key on each of its pages. The marking is a reconstruction from the commit history: the marked text is the model wording that history can attribute. Wording from earlier iterations of the report predates the records that would identify a model, and is left unmarked, as is wording introduced by any other means. Only passages of at least three consecutive lines are marked, so the copy shows blocks rather than scattered words. It is an estimate of origin, not a per-letter record of who typed what.

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Code and datasets are available at https://github.com/Dr1zz3l/spline-rio.

I. Introduction

Visual odometry and light detection and ranging (LiDAR)-based simultaneous localization and mapping (SLAM) are well-established, but both degrade in dusty, foggy, featureless, or textureless environments. Millimeter-wave (mmWave) frequency-modulated continuous-wave (FMCW) radar is robust to these conditions: it penetrates smoke and dust, operates in the dark, and measures Doppler velocity directly without feature matching. These properties make it attractive for aggressive flight in unstructured environments.

Combining a mmWave radar with an inertial measurement unit (IMU) (radar-inertial odometry (RIO)) for ego-motion estimation has been explored with discrete-time extended Kalman filter (EKF) and graph-optimization formulations [1], [2]. However, asynchronous radar-IMU measurements (radar at ∼10 Hz, IMU at ∼1 kHz) motivate a continuous-time trajectory, a smooth function evaluated at any sensor timestamp without interpolation or preintegration.

Research question. This work asks: *can a single-chip mmWave radar, fused with an IMU and a magnetometer, provide usable 6-DoF state estimation, and if so, what does it take?* The starting point is what the sensors can and cannot do: per-return Doppler yields direct ego-velocity, but at around a dozen returns per frame there aren't enough features for position fixing. The system is therefore a *velocity and orientation* source whose position estimate drifts with distance. Section V characterizes the radar measurements. The rest of the report develops an estimator informed by this characterization and evaluates it from gentle to aerobatic flight.

We contribute:

- a per-return characterization of a single-chip radar from hover to 10 rad/s, with every residual measured against motion capture ground truth: an uncertainty law with a per-return and a per-frame component, so one configuration is used for every evaluated flight and platform.

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- a continuous-time RIO pipeline based on that characterization: every radar return is an individually weighted Doppler factor in a B-spline estimator, with a consensus front end, sensor-only initialization, and a fixed-lag smoother with a marginalization prior, running in real time on a laptop CPU.
- an evaluation from gentle racing to aerobatic backflips, on held-out flights and on a public cross-platform dataset against published EKF baselines, including a consistency check of the reported covariance.

Data and code. The complete system is released for independent reproduction. The datasets and our Ceres solver with Python bindings are available at https://github.com/Dr1zz3l/spline-rio.

II. Related Work

A survey [3] reviews mmWave radar in robotic motion estimation, localization, and mapping. We focus on radar-inertial odometry for aerial platforms.

Radar ego-velocity estimation. Doppler-based ego-velocity estimation from mmWave radar was demonstrated by Kellner et al. [4] and extended to 3D with weighted least squares by Kramer et al. [2], whose RVE system fuses radar with IMU in a discrete-time EKF. Doer and Trommer [1] present an EKF-based RIO with online extrinsic calibration. Our work uses per-frame weighted least squares (WLS) ego-velocity estimates to initialize the position trajectory. The continuous-time sliding-window optimizer uses one Doppler factor per return (Section VI-E, Section VI-C).

Continuous-time trajectory estimation. Furgale et al. [5] introduced uniform B-splines for continuous-time visual-inertial calibration. The basalt visual-inertial odometry (VIO) [6] uses cumulative B-splines on Lie groups for stereo-inertial odometry; we adapt this basalt-style cumulative SO(3) B-spline to radar-inertial fusion.

Marginalization in sliding-window estimators. OKVIS [7] and VINS-Mono [8] use Schur complement marginalization for visual-inertial systems, propagating information from dropped keyframes as a dense Gaussian prior. We apply the same principle to continuous-time B-spline control points.

Continuous-time radar-inertial odometry. Ng et al. [9] formulate continuous-time radar-inertial odometry by fusing Doppler measurements from *multiple automotive* radars with an IMU as a least-squares problem over a B-spline trajectory, evaluated on passenger vehicles. Burnett et al. [10] use a Gaussian-process motion prior for spinning-radar- and lidar-inertial odometry, and Štironja et al. [11] model the IMU trajectory continuously inside an RIO estimator with online spatio-temporal calibration. Yang et al. [12] (Go-RIO) similarly fuse radar velocity with an IMU in continuous time and target the same radar elevation drift with a ground-plane model, which requires a persistently visible floor. Our aerobatic flights see it only intermittently. Our front end rejects Doppler outliers but does not impose a height constraint. Huang et al. [13] exploit radar cross-section and Doppler physics to make a sparse single-chip cloud usable in a discrete-time sliding-window optimization, evaluated on ground-robot and handheld platforms. We study a single-chip radar (10–60 points per frame, 12-channel array) on a quadrotor in aggressive racing and aerobatic backflip flight up to 10 rad/s. The radar measurement errors increase in this regime (Section IX-C).

Single-chip RIO on aerial platforms. Single-chip radar-inertial fusion has also been evaluated on multirotors. Michalczyk et al. [14] use a discrete-time EKF with scan-to-scan point matching and Doppler (a factor-graph reformulation follows in [15]); Girod et al. [16] run a real-time onboard M-estimator smoother on a quadrotor in cities and forests, but cap velocity at 2 m/s and anchor the vertical channel with a barometer. Our evaluation focuses on continuous-time estimation during aggressive and aerobatic flight, without a barometric height measurement (Section VIII-C). None of the systems above covers that combination.

Sliding-window marginalization and consistency. Sliding-window marginalization can introduce inconsistency in discrete time [17], where first-estimates Jacobian (FEJ) methods fix linearization points to preserve the observability nullspace. In continuous time, CLIC [18] and LIO-MARS [19] marginalize spline control points leaving the window by adjacency to the dropped control points. We follow the same construction and account for the globally shared IMU bias, which couples every IMU factor of the window and requires care when selecting factors for marginalization (Section VI-F).

Positioning against the EKF-RIO baselines. In the discrete-time UAV line, the EKF-RIO family of Doer and Trommer [1] provides our comparison baselines. Absolute root-mean-square error (RMSE) is not comparable across papers (firmware configuration alone changes the Doppler resolution by an order of magnitude), so we compare the systems on shared data: Section VIII-C runs our system on their public ICINS-2021 flights [20] and re-evaluates their estimators under our causal metrics on the same data.

III. Preliminaries

This section introduces the two mathematical tools the rest of the report builds on: uniform B-splines as a continuous-time trajectory representation, both in $\mathbb{R}^3$ and, in cumulative form, on $\mathrm{SO}(3)$; and the maximum-a-posteriori (MAP) estimator that fits them to the measurements as a weighted nonlinear least-squares problem.

A. Uniform B-splines in $\mathbb{R}^3$

An order-N (degree $N-1$) uniform B-spline with control points $\{\mathbf{P}_i\} \subset \mathbb{R}^3$ and constant knot spacing Δt represents a curve as a weighted sum of basis functions,

$$\mathbf{p}(t) = \sum_i \mathbf{P}_i \, B_{i,N}(t), \tag{1}$$

where the $B_{i,N}$ are the Cox–de Boor basis functions [5]. Two properties make this representation attractive for asynchronous sensor fusion.

Local support. At any instant only N consecutive control points have nonzero basis, so on the segment $t \in [t_i, t_{i+1})$,

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with normalized time $u = (t - t_i)/\Delta t \in [0, 1)$, the curve is a fixed blending of N control points,

$$\begin{aligned} \mathbf{p}(t) &= \sum_{j=0}^{N-1} B_j(u)\, \mathbf{P}_{i+j}, \\ \big[B_0(u) \;\cdots\; B_{N-1}(u)\big] &= \mathbf{u}^\top \mathbf{M}^{(N)}, \end{aligned} \tag{2}$$

with $\mathbf{u} = [1, u, \ldots, u^{N-1}]^\top$ and a constant $N \times N$ blending matrix $\mathbf{M}^{(N)}$ [6]. Local support limits each measurement's Jacobian to N control points and supports the marginalization described in Section VI-F.

Analytic derivatives. Differentiating the basis in (2) yields the velocity $\dot{\mathbf{p}}(t)$ and acceleration $\ddot{\mathbf{p}}(t)$ in closed form at *any* t, as lower-order splines over the same control points, with no finite differencing or inter-sample interpolation. This is what lets the accelerometer factor (which needs $\ddot{\mathbf{p}}$) and the min-snap regularizer (which needs $\mathbf{p}^{(4)}$) be evaluated exactly at each measurement's timestamp. We use a quintic (N=6) position spline so that snap is well defined and continuous.

B. Cumulative SO(3) B-splines

Orientation lives on the manifold SO(3), on which a weighted sum of rotations $\sum_i \mathbf{R}_i B_i$ is not itself a rotation, so (1) does not transfer directly. The *cumulative* parameterization of Sommer et al. [21], used in the basalt VIO [6], instead writes the orientation as a base rotation followed by a product of incremental rotations, each mapped back onto the manifold by the exponential map,

$$\begin{aligned} \mathbf{R}(t) &= \mathbf{R}_i \prod_{j=1}^{N-1} \mathrm{Exp}\big(\tilde{B}_j(u)\, \boldsymbol{\Omega}_j\big), \\ \boldsymbol{\Omega}_j &= \mathrm{Log}\big(\mathbf{R}_{i+j-1}^{-1} \mathbf{R}_{i+j}\big) \in \mathfrak{so}(3), \end{aligned} \tag{3}$$

where the increments $\boldsymbol{\Omega}_j$ are the relative rotations between consecutive *absolute* orientation control points $\mathbf{R}_i \in \mathrm{SO}(3)$ (the optimization variables, basalt convention). These orientation control points are also called "knots" in the continuous-time estimation literature [21]; here, we use "control points" for both coefficient sets and reserve "knots" for the time values defining the spline intervals. The cumulative basis is the tail sum of the ordinary basis of (2),

$$\tilde{B}_j(u) = \sum_{l=j}^{N-1} B_l(u). \tag{4}$$

Because each $\mathrm{Exp}(\cdot)$ returns a rotation, $\mathbf{R}(t) \in \mathrm{SO}(3)$ holds by construction for any orientation control points. This form retains absolute orientation control points, preserves local support over N control points exactly as in the $\mathbb{R}^3$ case, and yields the body angular velocity $\boldsymbol{\omega}_b(t)$ in closed form by differentiating the product, with no numerical differentiation. We use a cubic (N=4) orientation spline; (10) in Section VI-A is this construction with N=4 (base control point $\mathbf{R}_{k-3}$, three increments).

C. MAP Estimation as Weighted Least Squares

The control points and biases above form the finite-dimensional state $\mathcal{X}$ (Section IV-A). Each measurement $\mathbf{z}_i$ relates to the state through a known model $\mathbf{h}_i$, $\mathbf{z}_i = \mathbf{h}_i(\mathcal{X}) + \mathbf{n}_i$, with the noise $\mathbf{n}_i$ zero-mean Gaussian of covariance $\boldsymbol{\Sigma}_i$ and independent across measurements. The likelihood is then $p(\mathbf{z} \mid \mathcal{X}) \propto \prod_i \exp(-\frac{1}{2}\mathbf{r}_i^\top \boldsymbol{\Sigma}_i^{-1} \mathbf{r}_i)$ with residuals $\mathbf{r}_i = \mathbf{z}_i - \mathbf{h}_i(\mathcal{X})$, so its negative logarithm is a weighted sum of squared residuals: maximum-likelihood estimation minimizes $\frac{1}{2}\sum_i \mathbf{r}_i^\top \boldsymbol{\Sigma}_i^{-1} \mathbf{r}_i$, in which each residual is weighted by its inverse covariance (information) $\boldsymbol{\Sigma}_i^{-1}$. For a scalar residual this is a scalar weight $\lambda_i = 1/\sigma_i^2$: a more certain sensor is trusted more. The variance measurements in Section V inform the stream weights λ_a, λ_g of Section VI-B and the per-return radar weights of Section VI-C.

Prior knowledge $p(\mathcal{X})$, here trajectory smoothness and, online, the marginalization prior summarizing past data, turns the estimate into a MAP one, $\arg\max_{\mathcal{X}} p(\mathbf{z} \mid \mathcal{X})\, p(\mathcal{X})$; its negative log contributes the extra quadratic terms E_{reg} and E_{prior}. In a sliding window the prior is itself a posterior: the previous window's posterior over the states both windows share, with the states that left the window integrated out (Section VI-F). Radar Doppler residuals have heavy tails (Section V-F measures both the Gaussian core and the tails); the identifiable outliers are removed before the solve (Section V-B), so every residual that reaches the cost enters it quadratically. Collecting all terms gives the weighted nonlinear least-squares objective instantiated for our sensors in (6). We minimize it with Levenberg–Marquardt and analytic Jacobians. Rotation updates are applied on the manifold, $\mathbf{R} \mapsto \mathbf{R}\,\mathrm{Exp}(\boldsymbol{\delta})$, so every iterate remains a valid rotation. The Jacobians use spline values and derivatives at each measurement's exact timestamp, provided in closed form by Section III-A and Section III-B.

IV. System Overview

A. Problem Formulation

Estimation problem. We estimate the continuous-time 6-DoF body trajectory (world-frame position $\mathbf{p}_w(t) \in \mathbb{R}^3$ and orientation $\mathbf{R}(t) \in \mathrm{SO}(3)$) together with the constant IMU biases $\mathbf{b}_a, \mathbf{b}_g$, from asynchronous measurements over a sliding estimation window: radar Doppler velocities (∼11 Hz), accelerometer and gyroscope samples (1 kHz), and a low-rate heading prior. We adopt a *continuous-time* representation because the sensors are asynchronous and run at disparate rates: a trajectory defined for every t is evaluated at each measurement's exact timestamp, and its derivatives (velocity $\dot{\mathbf{p}}_w$, acceleration $\ddot{\mathbf{p}}_w$, and body angular velocity $\boldsymbol{\omega}_b$) follow in closed form, so no inter-sample interpolation or rigid-keyframe approximation is required.

Parameterization. The position and orientation spline control points (Section VI-A) and the IMU biases form the estimated state

$$\mathcal{X} = \big\{\, \mathbf{P}_i,\ \mathbf{R}_i,\ \mathbf{b}_a,\ \mathbf{b}_g \,\big\}. \tag{5}$$

The radar extrinsics are fixed at their measured values.

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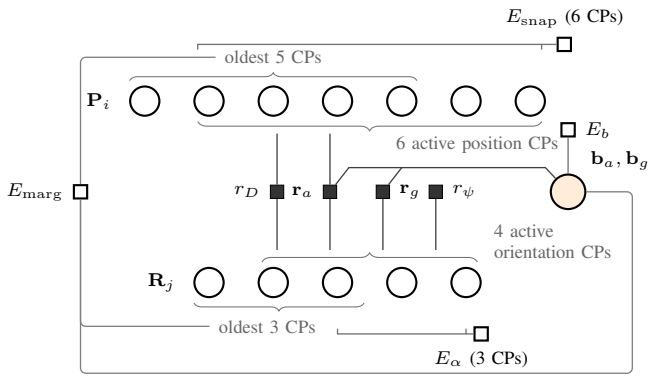


Fig. 1. Factor graph of one measurement time t. Circles are variables (position control points (CPs) $\mathbf{P}_i$, orientation control points $\mathbf{R}_j$, the shared IMU bias); filled squares are measurement factors, open squares the regularizers and the priors. Each factor at time t touches exactly the 6 active position CPs and/or the 4 active orientation CPs (braces): the radar Doppler factor r_D couples both splines (one such factor per return; the returns of a frame are stacked into one block, Section VI-C), the accelerometer $\mathbf{r}_a$ both splines plus the bias, the gyroscope $\mathbf{r}_g$ the orientation spline plus the bias, the heading factor r_ψ the orientation spline only. The minimum-snap regularizer E_{snap} spans 6 consecutive position CPs and the angular-acceleration regularizer E_α 3 consecutive orientation CPs; the bias prior E_b anchors the bias to its pre-flight value (Section VI-A); the marginalization prior E_{marg} binds the window's oldest $N-1$ control points of each spline and the bias (Section VI-F). Not drawn: the priors of the first window (Section VI-E). Each row runs forward in time, but the two grids differ (40 and 16 ms), so horizontal positions are not aligned across rows.

Estimator. Treating the measurement noise as zero-mean and independent, the MAP estimate is the minimizer of the negative log-posterior, a weighted nonlinear least-squares problem over the measurement residuals, the spline regularizers, and the priors, whose factor-graph structure Fig. 1 illustrates:

$$\begin{aligned}\hat{\mathcal{X}} = \arg\min_{\mathcal{X}} & \lambda_r \sum_k w_k\, r_{D,k}^2 + \lambda_a \sum_m \|r_{a,m}\|^2 \\ & + \lambda_g \sum_m \|r_{g,m}\|^2 + \lambda_\psi \sum_n r_{\psi,n}^2 \\ & + E_{\text{reg}}(\mathcal{X}) + E_{\text{prior}}(\mathcal{X}),\end{aligned} \tag{6}$$

where r_D, r_a, r_g, r_ψ are the radar-Doppler, accelerometer, gyroscope, and heading residuals of Section VI-B. There is one radar residual per return k, with w_k the return's own weight from the measured noise law. The returns of a frame are stacked and whitened for their shared error component (Section VI-C). The $\lambda_\bullet$ are the stream weights of Section V-G. E_{reg} collects the position min-snap and orientation smoothness regularizers (Section VI-D), and E_{prior} collects the bias prior, the window-start priors and, online, the marginalization prior E_{marg} (Section VI-F). We minimize (6) with Levenberg–Marquardt using analytic Jacobians. The fixed-lag estimator solves it over a 3 s window and, on each stride, marginalizes the states leaving the window into E_{marg} via the Schur complement (Section VI-F), keeping each window's problem size bounded.

B. Hardware Platform

The system runs on a quadrotor with a TI IWR6843AOPEVM mmWave radar (60–64 GHz FMCW), the flight controller's IMU, and a Vicon motion capture (MoCap) system used for ground truth evaluation. The radar is mounted upside-down; the extrinsic rotation is $[180°, 27.5°, 0°]$ (roll, pitch, yaw) and translation $[0.08, 0.02, -0.01]$ m in the body frame, with the 27.5° pitch the as-built downtilt measured by inclinometer (27–28° on the radar face). The radar operates at ∼11 Hz with 0.049 m/s Doppler resolution and the IMU at ∼993 Hz. Both are used at full rate.

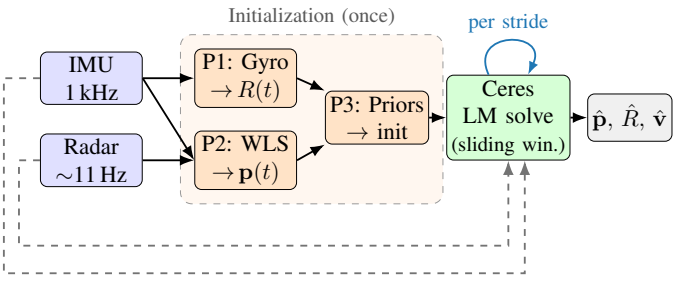


Fig. 2. System pipeline. Solid: initialization flow (P1–P3 seed the optimizer, run once). Dashed: raw sensor data fed as measurement factors. Blue: the sliding-window re-solve at each stride. Radar returns pass a RANSAC ego-velocity prefilter (Section VI-C) first. MoCap supplies only the yaw heading prior (magnetometer surrogate) and RMSE evaluation.

C. Pipeline

Fig. 2 shows the pipeline. A three-stage sensor-only initialization (P1–P3, detailed in Section VI-E) builds the initial trajectory the nonlinear solve needs, from radar and IMU data alone. The same raw measurements then also enter the solver directly as factors (Fig. 2, dashed). MoCap supplies only the yaw heading prior (a noise-free, idealized magnetometer surrogate, whose gap to a real sensor Section IX-B bounds) and the final RMSE evaluation. P1–P3 and the position/attitude estimate use no external pose reference.

V. Radar Measurement Characterization

This section characterizes the single-chip radar measurements used for state estimation: their timing, outliers, state-dependent noise, within-frame correlations, and residual distribution. These properties inform the estimator design in Section VI. Table I summarizes the measurements.

Radar residuals are evaluated against motion capture, independently of the estimator's output. The forward model in Section VI-B predicts each return's Doppler from the reference state. This prediction is used to unwrap aliased measurements and compute per-return residuals.

Two residual statistics are used throughout. The first is σ_{rob}, the median absolute deviation scaled by 1.4826, which estimates the standard deviation for Gaussian data [22]; it measures the width of the core while limiting sensitivity to outliers. The second is the *tail mass*, the fraction of residuals with magnitude greater than $3\sigma_{\text{rob}}$, approximately 0.27% for a zero-mean Gaussian.

A. Arrival Timing and Sparsity

The radar delivers frames at ∼10 Hz without a hardware trigger. Each frame is timestamped on arrival at the USB

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TABLE I
MEASURED RADAR DATA PROPERTIES (PER-RETURN DOPPLER RESIDUALS AGAINST THE MOCAP FORWARD MODEL, GROUND-TRUTH DE-ALIASED)

Property	Slow racing	Fast racing	Backflips
Returns per frame (median)	13	11	10
Frames below 5 returns	6%	12%	9%
Aliased returns	2.0%	23.7%	37.8%
Raw: σ_{rob} (m/s)	0.097	0.189	0.354
Raw: tail mass ($> 3\sigma_{\mathrm{rob}}$)	8.6%	17.8%	6.6%
Non-aliased: σ_{rob} (m/s)	0.094	0.138	0.279
Aliased: σ_{rob} (m/s)	0.62	0.72	0.52
After prefilter: σ_{rob} (m/s)	0.085	0.111	0.243
After prefilter: tail mass	3.1%	5.3%	2.1%
Law noise floor σ_0 (m/s)	0.049	0.140	0.327
Law bearing constant $\delta\phi_0$		2.7° (shared)	
Frame-shared term σ_c	0.96	1.38	1.55

Radar frame rate ∼10 Hz, IMU rate 993 Hz on all flights. "After prefilter" = the non-aliased returns kept by the consensus prefilter of Section V-B; aliased returns are handled separately. σ_0 and $\delta\phi_0$ from the per-return fit of Section V-C, used to determine the deployed constants (the binned fit drawn in Fig. 4 gives $\delta\phi_0 = 1.7°$); σ_c in units of the noise floor σ_0 (Section V-D).

port; individual returns have no separate timestamps. Frames contain a median of 10–13 returns, with 6–12% containing fewer than five (Table I), the minimum used for the consensus fit. The IMU runs asynchronously at 993 Hz.

Consequence. We use a continuous-time B-spline to evaluate the state and its derivatives at the asynchronous radar and IMU timestamps (Section VI-A). Sparse frames can make per-frame ego-velocity estimates unreliable, so the estimator uses individual Doppler returns instead of a fitted velocity measurement.

B. Outliers and the Prefilter

Some outliers can be identified within a frame, which is usually overdetermined (a median of 10–13 returns for a 3-DoF ego-velocity). A return whose Doppler disagrees with the velocity the other returns agree on can be rejected before optimization. We use reve's three-point RANSAC ego-velocity consensus filter [1] with the default inlier threshold of 0.15 m/s. The returns it rejects carry a median absolute residual several times that of the returns it keeps (7.4× on slow racing, 5.5× on fast racing, 2.5× on backflips, where the whole frame is noisy), and it keeps 81–91% of the non-aliased returns (Table I). Deleting the same number of returns at random removes almost no tail mass. A sweep of the inlier band from 0.05 to 0.6 m/s shows that tight bands retain too few returns in sparse frames, while loose bands retain outliers, with no value clearly better than the adopted 0.15 m/s.

Consequence. The consensus prefilter rejects outliers once per frame, before optimization. Section VIII evaluates its effect by removing it.

C. The Per-Return Noise Law

The residual core width varies with speed and return geometry. Binning the within-frame deviations of the non-aliased returns (the law's fit set: frames with at least five returns, 6σ-clipped) by platform speed $|\mathbf{v}|$, the angle θ_j between the ray and the velocity vector, and by the angle ϕ_j of the ray off the radar boresight (Fig. 4a–c) shows three trends on every flight: the noise grows linearly with speed, peaks for rays perpendicular to the motion, and increases off boresight, by 5.7, 2.3 and 4.3× between the innermost and outermost bins of Fig. 4c on slow racing, fast racing and backflips. We use the within-frame deviation $d_j = (r_j - \bar{r}_f)/\sqrt{1 - 1/N_f}$, where $\bar{r}_f$ is the mean residual of the N_f retained returns in frame f. Subtracting this mean removes an additive offset shared by all returns, but not ray-dependent reference or timing errors. The denominator corrects the variance reduction from mean subtraction for independent, equal-variance per-return errors; it is approximate when variances differ.

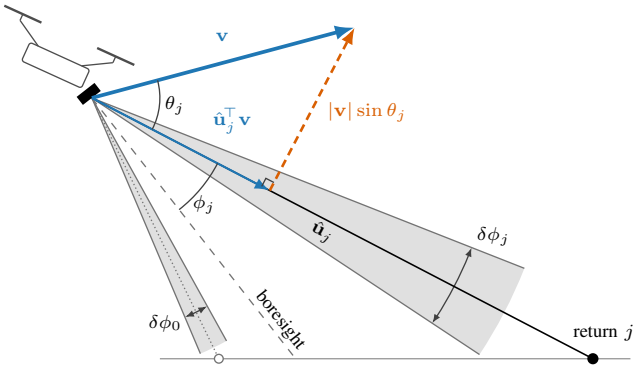


Fig. 3. Geometry behind the per-return law (Eq. (7)) in side view. The platform moves with velocity $\mathbf{v}$ past static scene points. Return j is the ray of direction $\hat{\mathbf{u}}_j$ from the radar to one of them. The sensor reads only the along-ray part of the velocity, the Doppler $v_D = -\hat{\mathbf{u}}_j^\top \mathbf{v}$ (positive for a receding target, so negative here). A bearing error swings the ray about the radar (grey cones) and leaves its length unchanged, so to first order it projects only the transverse velocity, of magnitude $|\mathbf{v}| \sin\theta_j$, into the Doppler residual. θ_j is the angle between the ray and $\mathbf{v}$. The angular width of the cone grows off the boresight (dashed: the sensor x axis, mounted 27.5° below the body's forward axis), from about $\delta\phi_0$ near the boresight to $\delta\phi_j = \delta\phi_0/\cos\phi_j$ at the off-boresight angle ϕ_j. This widening follows from the angle FFT's uniform binning in the direction cosine $\sin\phi$.

Derivation. Bearing error provides a model for these three trends. We derive it first in the angular form of Fig. 3. The deployed weight uses the direction-cosine form of the same law (Section VI-C). The sensor measures radial velocity along the *true* ray, $v_{\mathrm{meas}} = -\hat{\mathbf{u}}_{\mathrm{true}}^\top \mathbf{v}$, while the forward model predicts with the *reported* ray, $v_{\mathrm{pred}} = -\hat{\mathbf{u}}_{\mathrm{rep}}^\top \mathbf{v}$. The residual due to the direction error $\delta\hat{\mathbf{u}} = \hat{\mathbf{u}}_{\mathrm{rep}} - \hat{\mathbf{u}}_{\mathrm{true}}$ is $r_j = \delta\hat{\mathbf{u}}^\top \mathbf{v}$. Both rays are unit vectors, so to first order $\delta\hat{\mathbf{u}}$ is perpendicular to the ray, and only the transverse velocity $\mathbf{v}_\perp$, of magnitude $|\mathbf{v}| \sin\theta_j$, couples into the residual, $r_j \approx \delta\hat{\mathbf{u}}^\top \mathbf{v}_\perp$. For isotropic tangent-plane error with total angular rms $\delta\phi_j$ across its two components, the component along $\mathbf{v}_\perp$ has rms $\delta\phi_j/\sqrt{2}$. The angle FFT uses uniformly spaced bins in the *direction cosine* $\sin\phi$. Differentiating $\sin\phi$ converts a constant direction-cosine error into the angle error $\delta\phi_j = \delta\phi_0/\cos\phi_j$. The bearing-error contribution is therefore

$$\sigma_j = \frac{\delta\phi_0}{\sqrt{2}} \frac{|\mathbf{v}| \sin\theta_j}{\cos\phi_j}, \tag{7}$$

which captures the speed and geometry dependence. Each return also contains speed-independent noise, including detection and quantization noise. Independent sources add in

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variance, so with $q_j = |\mathbf{v}| \sin\theta_j / \cos\phi_j$ the total variance is modeled as

$$\sigma_j^2 = \sigma_0^2 + (c\, q_j)^2, \qquad c = \delta\phi_0/\sqrt{2}, \tag{8}$$

with a noise floor σ_0 and coefficient c. The deployed weight uses the direction-cosine form in (16) (Section VI-C), assigning constant error to the two resolved cosines and fixing the third by unit norm. The constants reported below are fitted to that form.

Fit and test. Fitted to the binned robust width of the per-return deviations on the two racing flights (10 quantile bins per flight, one shared coefficient and a per-flight floor), the law reproduces the binned widths with $R^2 = 0.98$. The normalized deviations d_j/σ_j provide a further check: their robust width should be approximately 1 across bins of each input (Fig. 4d). The full law gives approximately flat profiles; omitting a factor leaves a trend along the corresponding input. The transverse factor shows the smallest effect in Fig. 4d because most racing returns have $\sin\theta \geq 0.5$ (73–80%); it is retained because it follows from the bearing-error model and improves the fit ($R^2 = 0.98$ against 0.93 without). Held out of the fit, the backflip flight keeps this width near 1 along speed and θ, but excess noise remains off boresight. Additional terms fitted jointly to the same statistic receive zero coefficients, including the body-rate term. Thus body rate adds no explanatory value to this fit after accounting for speed and geometry.

Constants. Fitting the racing flights gives $\delta\phi_0 = 2.7°$. Adding the backflip flight with its own floor changes this estimate by 3%, supporting a shared coefficient. The firmware uses 64 direction-cosine bins over the field of view. The fitted error corresponds to approximately one bin per axis. The noise floor varies across regimes ($\sigma_0 = 0.049/0.140/0.327$ m/s on the three flights, Table I), collecting the speed-independent errors the law does not model. Possible sources include detection noise, smearing at high body rate, and errors near the aliasing boundary. A shared floor would confound these differences with the bearing-error coefficient. The solver uses the geometric mean of the two racing floors, $\sigma_0 = 0.083$ m/s, and Section VIII-E measures the sensitivity to that choice.

Consequence. Each return receives its own weight, $w_j = \sigma_0^2/\sigma_j^2 = 1/(1 + (c\, q_j/\sigma_0)^2)$, from its own geometry and the frame's speed; the weight halves at $q_j = \sigma_0/c = 2.5$ m/s. Speed is obtained from the frame's consensus ego-velocity. The IMU is used in the radar front end only for alias unwrapping (Section VI-C). The fitted model does not include a body-rate term.

D. Frame-Shared Error

The returns of one frame share a timestamp, an attitude and an integration interval, which can introduce correlated errors. We estimate a frame-shared component σ_c by separating within-frame variation from variation in the frame means. In units of the noise floor σ_0 of the deployed law, $\sigma_c = 0.96/1.38/1.55$ on slow racing, fast racing and backflips, and 0.83–1.02 on the four public ICINS flights (Section VIII-C): the shared component is of the same order as the per-return noise and increases across our three flight regimes. This correlation reduces the effective number of independent returns [23].

Consequence. Treating the returns of a frame as independent would overstate the information in their shared component. The estimator includes this component in the frame covariance and whitens the residual stack as described in Section VI-C. We use $\sigma_c = 1.27$, the geometric mean over our three flights, for every platform.

E. Aliasing

The firmware wraps Doppler into $\pm v_{\max} = \pm 3.14$ m/s. Measured against ground truth, 2.0% of returns are aliased on slow racing, 23.7% on fast racing and 37.8% on backflips. Unwrapping recovers the Doppler ambiguity but does not correct bearing errors. At matched speed, aliased returns have within-frame noise 2–4.6× that of non-aliased returns, with a noise floor of $\sigma_{\mathrm{al}} \approx 0.6$ m/s and weaker state dependence.

Consequence. The measurement consumed by the estimator is the *unwrapped* Doppler, selected once per frame against an IMU-integrated velocity prediction (Section VI-E). Selecting the correct wrap requires the predicted radial velocity to be within $v_{\max}$ of the true value. Aliased returns are retained with the reduced weight given by the measured variance ratio $w_{\mathrm{al}} = (\sigma_0/\sigma_{\mathrm{al}})^2 = 0.019$. Section VIII compares this choice with full weighting and deletion.

F. Doppler Residual Distribution

A zero-mean Gaussian error model with the specified covariance yields a covariance-weighted least-squares likelihood. The raw residuals have an approximately Gaussian core but heavy tails (Fig. 5a). The front end treats aliased returns separately (Section V-E) and rejects consensus outliers (Section V-B). Normalizing each retained return by its own σ_j (Section V-C) reduces the excess mass between 1.5σ and 3σ, while the prefilter reduces the far tail (Fig. 5b).

For the retained, non-aliased returns, the mass beyond 3σ is 3–10× the Gaussian value, down from 25–47× before the prefilter with one global σ. The remaining outlier population is not identified by the available flags: it accounts for 1–4% of the returns and is 5–10× wider than the core, against 6–12% and 6–11× before. Multipath is a possible source. The residuals remain approximately zero-mean and symmetric, with tails consistent with exponential decay over the observed range.

Consequence. We use a Gaussian likelihood with per-return variance σ_j^2 and frame-shared covariance (Section III-C, Section VI-C). This approximates the core but does not model the remaining heavy tails. A robust kernel after prefiltering produced no measurable improvement and is not used (Section VIII-E).

G. Stream Error Budgets

The noise law sets the relative weights of radar returns. The weights of the radar, gyroscope, and accelerometer streams must also account for temporal correlation. We approximate a stream sampled at f_s with integrated autocorrelation time τ (the sum of the residual autocorrelation over positive lags)

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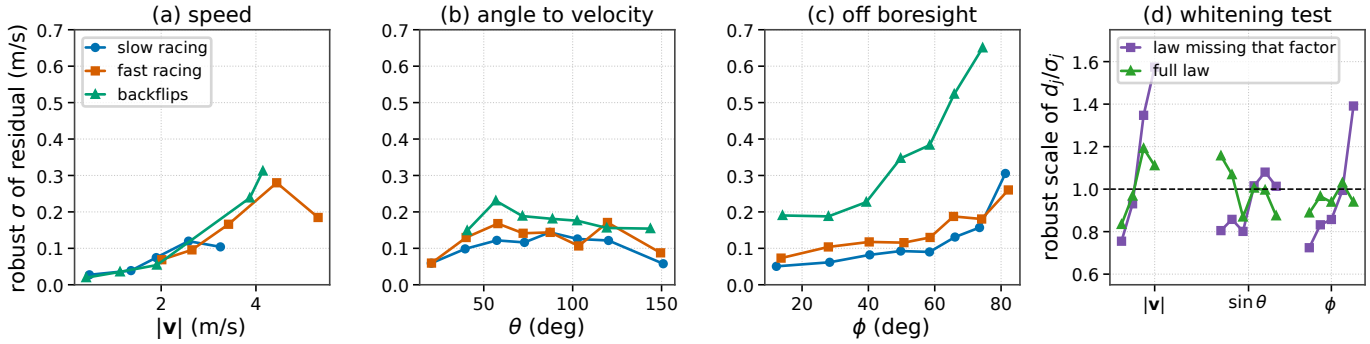


Fig. 4. The per-return noise law. (a–c) Robust width of the within-frame deviations binned by platform speed $|\mathbf{v}|$, by the angle θ between ray and velocity, and by the angle ϕ off boresight, for the three flights: the three trends of (7) are visible before fitting. (d) Model check on fast racing: robust scale of the normalized deviation d_j/σ_j binned along each input, for the full law (green) and the law missing the corresponding factor (purple). The expected scale is approximately 1; each missing factor leaves a trend along its own input.

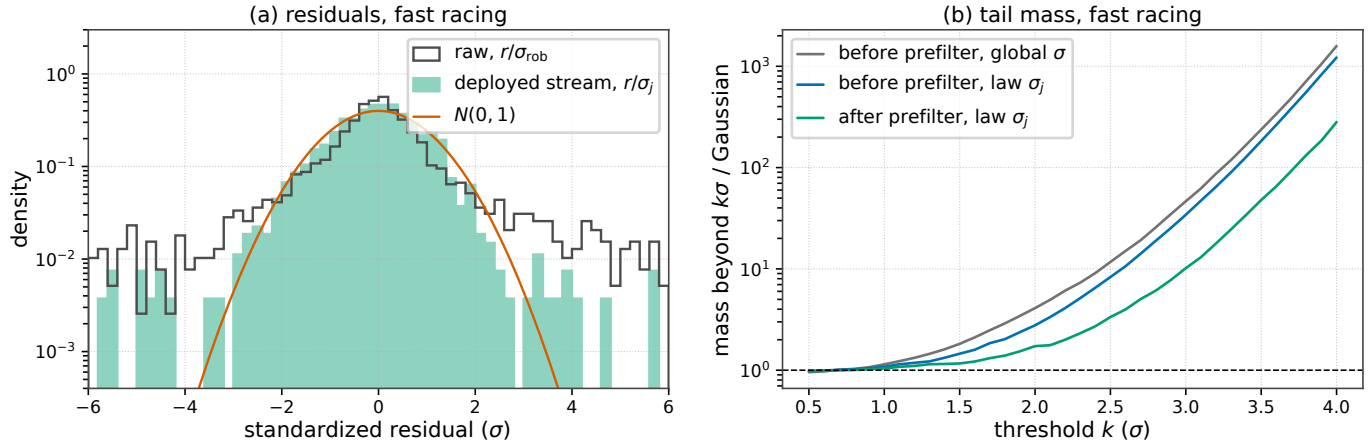


Fig. 5. The residual distribution before and after the front end (fast racing). (a) Standardized residuals on a log density scale: the raw stream in units of its σ_{rob} (grey) and the retained, non-aliased returns, each residual in units of its own σ_j (green), against $N(0,1)$. Aliased returns are weighted as their own population and are not in the green stream. (b) Tail mass beyond $k\sigma$ relative to a Gaussian, for the non-aliased returns before the prefilter with one global σ (grey) and with the per-return law σ_j (blue), and for the returns after the prefilter with the law (green). Together they reduce the excess by an order of magnitude. Per-return normalization mainly reduces the excess between 1.5σ and 3σ; the blue and grey curves converge beyond 3.5σ. The prefilter reduces the remaining far tail.

as providing $1/\tau$ independent samples per second. The corresponding per-sample weight is

$$\lambda^* = \frac{1}{\sigma^2 \, f_s \, \tau}, \tag{9}$$

which for white noise ($\tau = 1/f_s$) reduces to $1/\sigma^2$ per sample. Both inputs are measured per stream against the reference. For the IMU, we use residuals within the bandwidth represented by the splines. Much of the raw residual energy lies above this band: the gyroscope's in-band σ is 0.024–0.051 rad/s, compared with 0.15 rad/s before filtering. The measured IMU correlation times are 194–598 ms, corresponding to approximately 2–5 independent samples per second under this approximation. For the radar, σ is the root-mean-square residual of the non-aliased returns that pass the prefilter. Aliased returns have their own weight and are excluded from this estimate. The second moment includes the remaining outliers' contribution to the quadratic cost. Using the robust core width instead makes the weights from the two racing flights differ by a factor of 3.6.

Consequence. Taking the geometric mean of (9) over the two racing flights gives $\lambda_g = 3.13$, $\lambda_a = 0.0039$, and a radar weight of 2.64 per return (Section VI-C). Ignoring temporal correlation would increase the IMU weights by $f_s\tau$, a factor of several hundred, relative to this approximation. Section VIII-D tests the consistency of the resulting estimator covariance.

VI. Methodology

A. State Representation

Position. The world-frame position $\mathbf{p}_w(t) \in \mathbb{R}^3$ is a quintic (order-6) uniform B-spline with knot spacing $\Delta t_p = 40$ ms. Velocity $\dot{\mathbf{p}}_w$ and acceleration $\ddot{\mathbf{p}}_w$ are obtained as analytical derivatives of the spline.

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Orientation. The body-to-world rotation $\mathbf{R}(t) = \mathbf{R}_{wb}(t)$ is a cubic cumulative $\mathrm{SO}(3)$ B-spline [6], [21]:

$$\mathbf{R}(t) = \mathbf{R}_{k-3} \prod_{j=k-2}^{k} \mathrm{Exp}\big(\tilde{B}_j(t)\,\mathrm{Log}(\mathbf{R}_{j-1}^{-1}\mathbf{R}_j)\big), \tag{10}$$

where $\mathbf{R}_{k-3}, \ldots, \mathbf{R}_k$ are the four active orientation control points and $\tilde{B}_j(t)$ are the cumulative basis functions defined in Section III-B. The absolute orientation control points are optimized. The knot spacing is $\Delta t_o = 16\,\mathrm{ms}$, and the body-frame angular velocity $\boldsymbol{\omega}_b(t)$ follows by analytic differentiation. Both spline spacings are fixed across flights and platforms (Section IX-H).

Biases and extrinsics. The constant six-component IMU bias $[\mathbf{b}_a, \mathbf{b}_g]$ is estimated jointly with the trajectory. A Gaussian prior anchors it to the stationary initialization (Section VI-E), with standard deviations $0.216\,\mathrm{m/s^2}$ and $0.0036\,\mathrm{rad/s}$ for the accelerometer and gyroscope, respectively. These widths are measured offline from stationary ground segments, the accelerometer's against the specific force $-\mathbf{R}_{bw}\mathbf{g}_w$ predicted from the MoCap attitude, the gyroscope's from the bias shift between motors off and motors running. The prior weights are inverse variances, and the prior touches no leaving state, so it is re-applied in every window and never absorbed into the marginalization prior (Section VI-F). All radar extrinsics are fixed at the platform's measured values. On our platform, freeing the pitch left the estimate close to its initial value, so we use the measured $27.5°$ mount angle (Section IV).

B. Measurement Models

Radar Doppler with lever arm. For a static target at sensor-frame bearing $\hat{\mathbf{u}}_s$, the predicted Doppler is

$$v_D = -\hat{\mathbf{u}}_b^{\top}\left(\mathbf{R}_{bw}\,\mathbf{v}_w(t) + \boldsymbol{\omega}_b(t) \times \mathbf{t}_{bs}\right), \tag{11}$$

where $\mathbf{v}_w = \dot{\mathbf{p}}_w$, $\hat{\mathbf{u}}_b = \mathbf{R}_{bs}\hat{\mathbf{u}}_s$, $\mathbf{R}_{bs}$ rotates sensor coordinates into the body frame, and $\mathbf{R}_{bw} = \mathbf{R}_{wb}^{\top}$. The lever arm $\mathbf{t}_{bs}$ runs from the body center of mass (CoM) to the radar antenna, expressed in body coordinates. The sign follows the sensor convention: positive Doppler denotes a receding target. Each return contributes a residual $r_{f,j} = v_{D,\mathrm{meas}} - v_{D,\mathrm{pred}}$ to its frame's jointly weighted residual block (Section VI-C).

Accelerometer.

$$\mathbf{r}_a = \mathbf{z}_a - \mathbf{R}_{bw}(\ddot{\mathbf{p}}_w - \mathbf{g}_w) - \mathbf{b}_a, \tag{12}$$

where $\mathbf{z}_a$ is the measured specific force and $\mathbf{g}_w$ is gravity in world coordinates.

Gyroscope.

$$\mathbf{r}_g = \mathbf{z}_g - \boldsymbol{\omega}_b(t) - \mathbf{b}_g, \tag{13}$$

Here $\mathbf{z}_g$ is the measured body-frame angular rate. The weights for both IMU residuals are specified in Section VI-C.

Heading prior. Absolute yaw is unobservable from Doppler and IMU measurements: a common rotation of position, velocity and attitude about gravity leaves these measurements unchanged. We supply heading using MoCap yaw at approximately 100 Hz as an idealized magnetometer surrogate. With $\mathbf{h} = \mathbf{R}\,\mathbf{e}_x$ the body forward axis in world coordinates, the residual is

$$r_\psi = h_x \sin\psi_{\mathrm{ref}} - h_y \cos\psi_{\mathrm{ref}}, \tag{14}$$

i.e. $\sin(\psi - \psi_{\mathrm{ref}})$ scaled by the horizontal projection of the forward axis, with weight $\lambda_\psi = 0.6$. It needs no angle wrapping, vanishes when the nose points vertically (the heading is momentarily undefined through a flip), and has a spurious zero at the opposite heading, which the initialization avoids. Section IX-B discusses this weight and the limitations of the surrogate.

C. Radar Front End and Measurement Weighting

The radar front end unwraps Doppler, rejects consensus outliers and assigns weights before optimization. The measurement characterization in Section V motivates these steps.

Alias unwrapping and the alias weight. At frame load, each return's Doppler alias is selected using the radial velocity predicted from IMU integration (Section VI-E). Correct selection requires the predicted radial velocity to be within $v_{\max}$ of the true value. All subsequent stages use this unwrapped Doppler. Unwrapping does not correct the bearing errors of aliased returns. These returns retain their reported bearings but receive the reduced weight $w_{\mathrm{al}} = 0.019$, the measured ratio of clean to aliased noise-floor variances (Section V-E).

Consensus prefilter. Following reve [1], three-point RANSAC estimates ego-velocity for each frame. Only returns within $0.15\,\mathrm{m/s}$ of the consensus prediction reach the solver (Section V-B). Frames with fewer than five returns bypass this filter. Filtering runs once at frame load.

Per-return weights in the frame covariance. For each retained, non-aliased return, the relative inverse-variance weight from (8) is, in the simplified angular model,

$$w_{f,j} = \frac{1}{1 + (c\,q_{f,j}/\sigma_0)^2}, \qquad q_{f,j} = \frac{\|\mathbf{v}_f\| \sin\theta_j}{\cos\phi_j}, \tag{15}$$

where θ_j is the angle between velocity and ray, and ϕ_j is the angle off boresight. The constants are $c = \delta\phi_0/\sqrt{2}$, $\delta\phi_0 = 2.7°$ (converted to radians), and $\sigma_0 = 0.083\,\mathrm{m/s}$ (Section V-C). The solver uses the direction-cosine form of the same law: with $\mathbf{s}$ the unit bearing and $\mathbf{v}_s$ the frame velocity, both in sensor coordinates, and $c_\phi = \max(s_x, 0.15)$,

$$\begin{aligned} X_j &= \sqrt{(v_{s,y} - s_y v_{s,x}/c_\phi)^2 + (v_{s,z} - s_z v_{s,x}/c_\phi)^2}, \\ \sigma_j^2 &= \sigma_0^2 + (c\,X_j)^2, \end{aligned} \tag{16}$$

which needs no clamp towards $\phi = 90°$ beyond the floor on c_ϕ; $w_{f,j} = \sigma_0^2/\sigma_j^2$. The velocity $\mathbf{v}_f$ comes from the frame's WLS consensus fit. Weights are fixed when the problem is built, independently of later trajectory updates. They still depend on the IMU prediction through the earlier alias decisions.

The residuals of frame f are stacked in $\mathbf{r}_f$. Their covariance model is

$$\Sigma_f = \sigma_0^2\left[\mathrm{diag}(w_{f,j}^{-1}) + \sigma_c^2 \mathbf{1}\mathbf{1}^{\top}\right], \tag{17}$$

using $w_{f,j} = w_{\mathrm{al}}$ for aliased returns and $\sigma_c = 1.27$ for the shared component (Section V-D). The rank-one term accounts for a common additive error within a frame, reducing

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the information attributed to repeated observations of that component. The frame is weighted jointly, as specified below.

Accelerometer gate. The accelerometer weight decreases with body-rate magnitude to limit the influence of vibration and other unmodeled high-dynamic errors. The gate in (19) uses an empirical scale $\omega_a = 8\,\text{rad/s}$, chosen from the velocity–orientation trade-off. Translational acceleration itself is already included in (12). The gyroscope weight remains constant.

Static stream weights. The per-sample stream weights follow the correlation-adjusted error budgets of (9). Taking their geometric means over the two racing flights gives $\lambda_g = 3.13$, $\lambda_a = 0.0039$ and $\lambda_r = 2.64$. These weights are fixed across flights and platforms.

The complete weighting. The effective information weights are

$$\Lambda^{\text{gyro}} = \lambda_g, \tag{18}$$

$$\Lambda^{\text{accel}} = \frac{\lambda_a}{1 + (\|\boldsymbol{\omega}_b\|/\omega_a)^2}, \tag{19}$$

$$\Lambda_f^{\text{radar}} = \lambda_r\, \sigma_0^2 \Sigma_f^{-1}. \tag{20}$$

Thus the radar contribution is $\sum_f \mathbf{r}_f^\top \Lambda_f^{\text{radar}} \mathbf{r}_f$. The factor σ_0^2 normalizes the covariance because λ_r already includes the stream's inverse-variance scale. Without within-frame correlation, this reduces to $\lambda_r \sum_{f,j} w_{f,j} r_{f,j}^2$. All retained measurement residuals enter quadratically, without a robust kernel. The constants in these weights are measured except for the empirical accelerometer gate.

D. Regularization

Position: minimum snap. $E_{\text{snap}} = \lambda_s \sum_i \|\mathbf{p}^{(4)}(t_i + \Delta t_p/2)\|^2$, one term per knot interval (the midpoint sum of the snap integral, without the Δt_p factor), with $\lambda_s = 2{\times}10^{-5}$.

Orientation: angular-increment smoothness. We penalize changes between consecutive rotation increments:

$$\mathbf{r}_\alpha = \text{Log}(\mathbf{R}_{i-1}^{-1}\mathbf{R}_i) - \text{Log}(\mathbf{R}_i^{-1}\mathbf{R}_{i+1}), \tag{21}$$

The cost is $\lambda_\alpha \sum_i \|\mathbf{r}_{\alpha,i}\|^2$ with $\lambda_\alpha = 0.1$. This discrete smoothness penalty is zero for equal consecutive increments. It approximates angular-acceleration regularization at the fixed knot spacing but is not a time-normalized angular-acceleration residual. Both regularization weights are design choices (Section IX-F).

E. Sensor-Only Initialization (P1–P3)

P1: Orientation. An automatically detected stationary segment provides the mean gyroscope bias and a gravity-aligned scalar accelerometer correction. This does not identify the full accelerometer bias independently of tilt. Initial roll and pitch follow from the corrected gravity direction, and yaw is set to zero until heading aiding is applied. Orientation is initialized by integrating debiased gyroscope readings: $\mathbf{R}(t + \Delta t) = \mathbf{R}(t)\,\text{Exp}\big((\mathbf{z}_g - \mathbf{b}_g)\Delta t\big)$.

P2: Position. The front end of Section VI-C produces a per-frame WLS ego-velocity estimate from the unwrapped, prefiltered returns. Rotated into the world frame with the initialized attitude (the lever arm is neglected at this stage), it is integrated with forward Euler into the initial position trajectory. The alias predictor of Section VI-C is the IMU-integrated velocity from the same attitude, and a frame whose consensus fails keeps all its returns.

P3: Boundary priors. Position, velocity and orientation priors anchor the first window at $t = 0$ to the initialized state, with hand-set weight $\lambda_{\text{bnd}} = 1000$. They are not re-applied in later windows, which carry previous information through the marginalization prior (Section VI-F).

F. Sliding Window with Marginalization

The fixed-lag smoother advances a 3 s window in 0.3 s strides. After each solve, position and orientation control points leaving the window are eliminated from the linearized problem by a Schur complement. The retained variables coupled to these states, including the shared bias, form the boundary of the next window. The resulting information matrix is stored as a Gaussian prior, with the approximations described below.

Which factors enter the marginal. Factors touching a leaving variable, including the previous prior when applicable, enter the Schur complement and are then removed from the active problem. Factors involving only retained variables remain active and must not also be absorbed into the prior. In particular, sharing the retained IMU bias does not justify including every IMU factor. The boundary is the last five position control points and three orientation control points before the window start, plus the bias (30 coordinates). A factor touching a leaving variable whose support would reach past this boundary, a sliver at the mismatch of the two grids, is dropped rather than absorbed, so the prior stays a marginal over exactly that boundary.

Re-centered prior. Before each solve, the prior is centered at the current warm start $\mathbf{x}_0$, retaining its information matrix and setting its gradient there to zero. This is a curvature-only approximation to the marginalized problem. It generally changes the prior's mean, so it does not preserve the full Gaussian marginal. The prior still penalizes subsequent departures from the warm start during the solve.

Staleness of the prior. To reduce reliance on older linearizations during rotation, the prior's position and orientation uncertainties are increased with accumulated angular travel. The update adds the variance $(k\theta)^2$ to every boundary position and orientation coordinate of the marginal's covariance (a Woodbury update, so the cross-covariances follow), with θ the rotation accumulated over the marginalized stride and the rates $k = 0.12\,\text{m}$ and $0.01\,\text{rad}$ per radian measured as the live-edge position and orientation error increments per swept radian against ground truth (the position rate is 0.12, 0.12 and 0.17 m/rad on the three flights). Angular travel is a proxy for linearization error, not a direct measure of it. Apart from this inflation, no global scale factor is applied to the prior. Section VIII-E evaluates the effect of position inflation, and Section VIII-D tests the output covariance.

Live-edge warm start. The incoming trajectory segment is aligned to the previous solution at its last data-supported

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TABLE II
DATASET SUMMARY

Flight	Dur. (s)	$v_{\max}$ (m/s)	$\lvert\omega\rvert_{\max}$ (rad/s)	Frames
Slow racing	25.8	3.65	2.62	258
Fast racing	18.0	5.66	9.14	179
Backflips*	18.1	∼4.3	∼10.9	181

*Backflip speed and angular rate are 95th percentiles rather than maxima because MoCap tracking artifacts dominate the instantaneous peaks.

position and orientation control points. This reduces mismatch at the transition before optimization.

VII. EXPERIMENTAL SETUP

Datasets. Table II summarizes the three flights used for characterization and the main evaluation, recorded as rosbags in a Vicon MoCap lab on the platform described in Section IV. Additional held-out flights (Section VIII-B) and four public ICINS-2021 flights with VINS pseudo ground truth (Section VIII-C) test transfer beyond these recordings.

Configuration. Unless varied in an ablation, all flights on both platforms use the configuration of Section VI: a 3 s window, 0.3 s stride, and knot spacings $\Delta t_p = 40$ ms and $\Delta t_o = 16$ ms, with no per-flight retuning. Radar extrinsics are fixed at each platform's calibration, measured on ours and published with the public dataset. Nothing else was re-measured for the second platform: the noise-model constants and stream weights are those measured on our own sensor and flights (Section V); the per-sample IMU weights were applied unchanged at its 409 Hz rate, and Section VIII-C reports what that costs. Initialization uses radar and IMU alone; heading aiding uses reference yaw as an idealized magnetometer surrogate with $\lambda_\psi = 0.6$ (Section IX-B).

Evaluation. The reference is MoCap on our flights and the public dataset's pseudo ground truth on theirs. Estimates are aligned to the reference by the rigid transform at the first frame only, without scale correction or a whole-trajectory fit. Such a fit is common in odometry work and is the weaker protocol, since it can absorb accumulated drift into the transform. The final 3 s is excluded from evaluation. Velocity RMSE compares the spline's analytical derivative with the low-pass filtered reference velocity. *Position drift* denotes position RMSE divided by the reference path length, expressed as a percentage. The *live* RMSE uses each window's newest stride before subsequent refinement; the *settled* RMSE uses the trajectory after all windows have refined it. For comparison with published results, Table IV(b) additionally reports the whole-trajectory absolute trajectory error (ATE) under Umeyama alignment.

VIII. RESULTS

A. Live-Edge Results

Table III reports live-edge and settled errors for the three flights using the configuration in Section VII. Fig. 6 and Fig. 7 show the trajectories and live-edge errors. We report velocity first because Doppler constrains it directly. Position is its integral.

TABLE III
SLIDING WINDOW: LIVE LEADING-EDGE AND SETTLED RMSE
(PER-WINDOW SOLVE TIME ON A LAPTOP CPU)

Flight	Mode	Vel (m/s)	Ori (°)	Drift (%)	t_{win} (s)
Slow racing	**Live edge**	**0.16**	**1.65**	**0.32**	0.16
	Settled	0.17*	1.16	0.29	
Fast racing	**Live edge**	**0.28**	**2.56**	**1.33**	0.18
	Settled	0.42*	2.41	1.28	
Backflips	**Live edge**	**1.66**	**12.1**	**2.91**	0.20
	Settled	1.72*	7.91	2.75	

The **bold** live-edge rows are the deployment-relevant figures. *The settled trajectory combines control points last optimized in different windows. Disagreement between those solutions distorts its velocity near stride boundaries and inflates the settled velocity RMSE, while live velocity comes from each window's own spline. Path lengths 48.1 / 44.9 / 54.1 m. Repeated solver runs on the same recordings vary by <2%.

On the racing flights, velocity RMSE is 0.16–0.28 m/s and position drift is 0.3–1.3% of the path length. Errors increase on the backflip flight, with orientation RMSE reaching 12.1° (Section IX-C). All three regimes solve faster than the 0.3 s stride on a laptop CPU (Table III), i.e. the deployed settings run in real time.

B. Held-Out Validation

The per-return law and the stream weights were fitted on the two racing flights with the backflip flight held out. The frame-shared term, the alias floor, the bias-prior widths and the staleness rates were measured on all three. The configuration was then run *unmodified* on flights never used for any decision.

A second, independent fast-racing flight was recorded with the radar's on-chip clustering disabled (61 returns per frame instead of 11). The front end then collapses duplicate detections to the physical cluster count, a preprocessing step that applies only when on-chip clustering is off, with the solver configuration unchanged. Velocity RMSE is 0.27 m/s, compared with 0.28 m/s on the fitting flight. Orientation RMSE is 2.7° and drift is 1.2%. The four public ICINS flights in Section VIII-C test transfer to a different platform.

Accuracy degrades on a held-out circular flight at 5 m/s. With the radar's Doppler ambiguity of 3.1 m/s, 63.8% of its returns must be alias-unwrapped, the IMU-integrated unwrapper's alias decisions degrade, and the live edge reaches 1.30 m/s / 11.7° / 9.1%. Down-weighting aliased returns cannot compensate for incorrect wrap selection.

C. External Cross-Validation Against EKF Baselines

We cross-validate against the EKF-RIO family of Doer and Trommer [1], [20] in both directions on shared data, with their toolbox pinned at a fixed commit and every parameter deviation documented in the repository.

Our system on their data. We run our solver on the four public ICINS-2021 indoor flights [20] (the same radar chip, a different IMU at 409 Hz, VINS pseudo ground truth). We use their published extrinsic calibration, *all other parameters remain those measured on our sensor and flights*; their own frame-shared term is $\sigma_c = 0.93$, and we kept 1.27. Their estimators (ekf-rio with unaided yaw and ekf-yrio with radar

Attributable AI wording: Fable 5.1 Opus 5

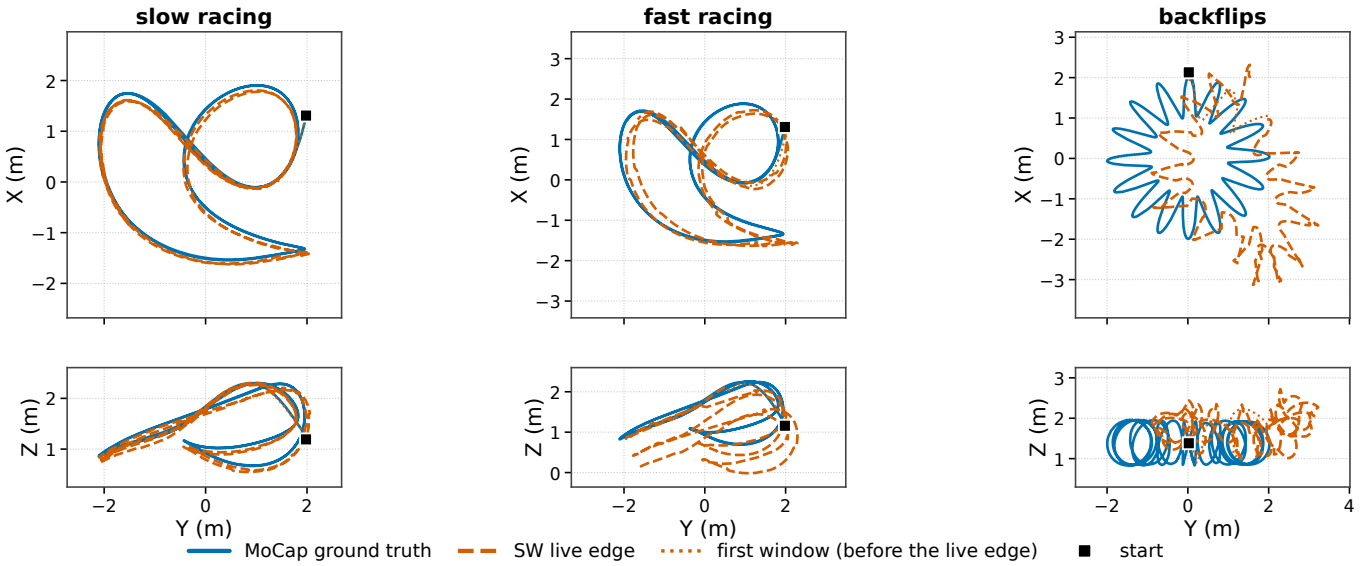


Fig. 6. Trajectory overlays for all three flights (columns): MoCap ground truth vs. the sliding-window live edge, X–Y (top) and Y–Z (bottom) projections. Within each column both projections share a common Y range and metric scale (equal aspect, so loops keep their true shape). The scale differs between columns. Start ■. the dotted lead-in is the first window's estimate, available only once that window has been solved (the smoother needs one complete 3 s window), so it is not a live output at those timestamps. It starts on the ground truth by the start-anchored alignment. The backflip flight circles in the horizontal plane while looping in the vertical plane. Its position drifts through the flips (2.9% of the path) while the loops remain visible in the orientation estimate (Section IX-C).

yaw aiding) are re-run from their stock configurations and evaluated with the same causal protocol. Table IV shows comparable velocity errors on every flight. Drift is 0.4–1.3% of the path, ahead of the unaided baseline on flight 1 and behind both baselines on the other three, with the difference in the vertical channel (Table IV(b), Section IX-E). The orientation column prefers our solver only because we anchor heading with a pseudo-magnetometer while theirs is sensor-only or unaided. On the heading-independent roll and pitch the three systems are similar (Table IV(b)), and with our heading aiding removed our yaw drifts similarly to their unaided filter (Section IX-B).

The consensus prefilter reduces our vertical drift on all four public flights. On flight 1 it falls from 14.3% to 0.35% of the path, an error the baselines do not show.

Their filters on our data. Ported unmodified onto our data, their filters, expecting order-100 clouds, fail on our sparse radar frames. On slow racing the $\chi^2(3)$ gate of ekf-rio rejects 99.7% of the radar ego-velocity updates, and the filter reverts to IMU dead-reckoning.

TABLE IV
CROSS-VALIDATION ON THE PUBLIC ICINS-2021 FLIGHT DATASETS [20], ALL SYSTEMS UNDER OUR CAUSAL PROTOCOL

(a) velocity [m/s] / orientation [°] / drift [% path]

Flight (path)	ekf-rio	ekf-yrio	Ours
1 (142 m)	0.20 / 10.1 / 1.0	**0.08** / 0.70 / **0.3**	0.09 / **0.55** / 0.4
2 (45 m)	**0.08** / 1.59 / 0.6	**0.08** / **0.91** / **0.4**	**0.08** / 1.00 / 1.0
3 (147 m)	**0.08** / 1.07 / **0.3**	**0.08** / 1.09 / **0.3**	**0.08** / **0.53** / 1.3
4 (78 m)	0.09 / 3.67 / 0.5	**0.07** / 1.41 / **0.3**	0.08 / **1.21** / 1.2

(b) Decomposition (systems: ekf-rio / ekf-yrio / Ours)

	Roll/pitch [°]			Our drift [%]		Whole ATE [m]		
Fl.	rio	yrio	Ours	hor.	ver.	rio	yrio	Ours
1	0.24	0.25	0.22	0.21	0.35	0.87	0.23	0.41
2	0.57	0.59	0.58	0.55	0.85	0.13	0.10	0.22
3	0.25	0.26	0.24	0.49	1.28	0.36	0.26	0.90
4	0.74	0.74	0.75	0.61	1.08	0.30	0.15	0.49

Live estimate, start-anchored alignment. **(a)** headline, with heading aiding differing by design (see text). **(b)** decomposition: *roll/pitch* is the heading-independent attitude RMSE (yaw excluded), so the orientation differences in (a) mainly reflect yaw. *Our drift* splits position into horizontal/vertical. *Whole-traj. ATE* re-evaluates every system under EVO/KITTI Umeyama alignment (no scale) for cross-checking published figures. Prefilter *disabled*, flight 1: vertical drift 14.3%, whole-traj. ATE 11.7 m.

D. Output Covariance Consistency (NEES)

The reported live-edge covariance of $(\mathbf{v}_w, \mathbf{R})$ is tested with the normalized estimation error squared (NEES) [24]. Each window's error against ground truth is normalized by that covariance. For a consistent estimator each 3-DoF block has mean 3. On both racing flights the reported covariance is *conservative*: the velocity NEES mean is 0.20 on slow racing (N=49) and 0.43 on fast racing (N=20), the orientation mean 0.39 and 1.04, against the expected 3 (nominal 95% χ^2 intervals for the N-window mean, $[2.4, 3.7]$ and $[2.0, 4.2]$, which treat the overlapping windows as independent). The reported standard deviations are thus 1.7–3.8× too large. On backflips only 6 of 49 windows remain after the ground-truth mask. Their covariance is overconfident (1.4× in velocity and 3.2× in orientation, in standard deviation), but the sample is too small for a firm conclusion.

E. Ablations

Attributable AI wording: Fable 5.1 — Opus 5

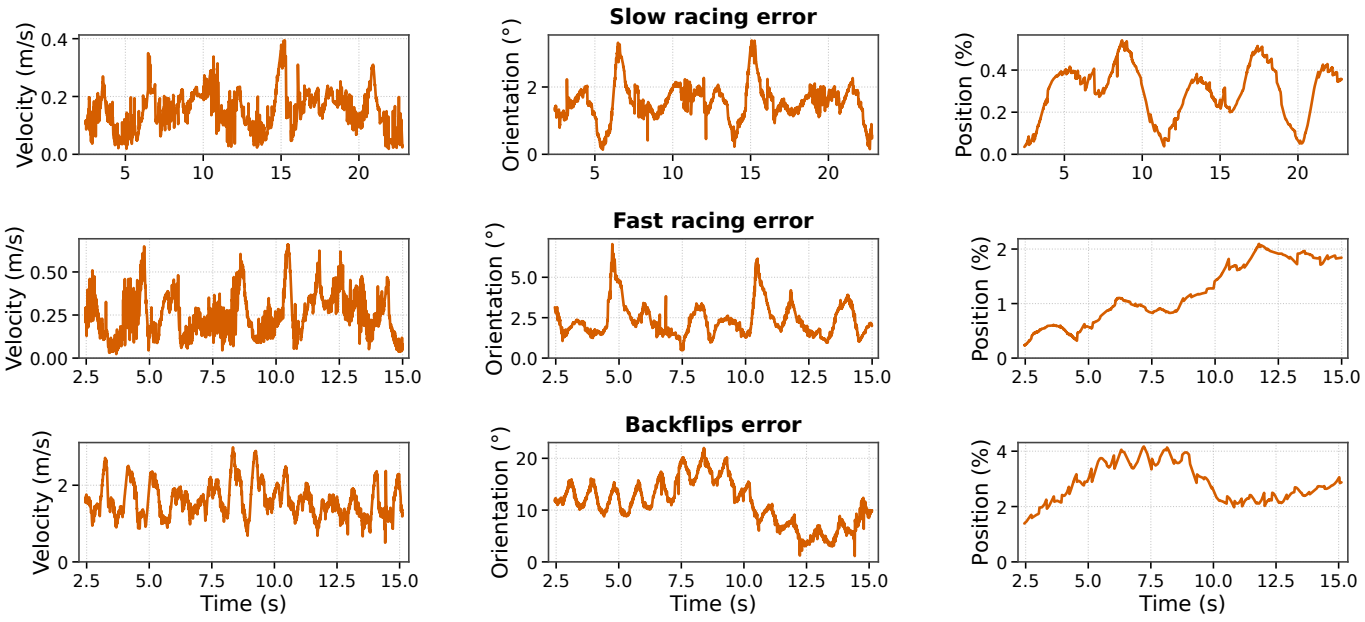


Fig. 7. Live-edge velocity, orientation, and position error (the last in percent of the flight's path length, the unit of the drift metric) vs. time for slow racing (top), fast racing (middle), and backflips (bottom). Velocity and orientation errors spike at high-speed, high-dynamics sections. Fast-racing position error accumulates with trajectory length. Backflip errors grow step-wise *during* the flips, where the radar measurement model degrades (Section IX-C).

TABLE V
SENSITIVITY TO THE LAW'S NOISE FLOOR σ_0 (LIVE EDGE: VELOCITY / ORIENTATION / DRIFT)

σ_0	Slow racing	Fast racing	Backflips
0.5× (0.041 m/s)	0.18 / 1.65 / 0.48	0.29 / 2.55 / 1.40	1.74 / 12.2 / 5.1
1× **(0.083, deployed)**	**0.16 / 1.65 / 0.32**	**0.28 / 2.56 / 1.33**	**1.66 / 12.1 / 2.9**
2.5× (0.207 m/s)	0.16 / 1.67 / 0.30	0.29 / 2.58 / 1.40	1.65 / 12.2 / 2.0

Selected front-end, weighting and prior choices are reverted one at a time at the deployed operating point (live edge: velocity / orientation / drift).

The noise floor σ_0. We use the noise floor derived from the racing fits on all flights (Section V-C). Table V tests values of 0.5 and 2.5 times this setting. On the racing flights, velocity error changes by about 7% (0.165 to 0.177 m/s on slow racing). On backflips, increasing the floor to 0.207 m/s reduces drift from 2.9 to 2.0%, with little change in orientation error, motivating the in-flight noise-floor estimate listed as future work.

Front end. Table VI reverts the front-end elements one at a time. The consensus prefilter is needed on the racing flights (velocity error +21% and +51% without it) and on the public dataset (Section VIII-C), where the vertical channel fails without it. On backflips, disabling the prefilter reduces drift from 2.9% to 2.4%. The filter removes returns from already sparse backflip frames. We accept this drift penalty to retain its benefits on racing and the public dataset. Robust kernels (radar $\delta = 1$ m/s, accelerometer $\delta_a = 2\,\mathrm{m/s^2}$) have negligible effect on slow racing. On fast racing, they slightly worsen velocity and orientation errors while reducing drift by 0.2 pp. With both enabled, backflip orientation error falls to 7.4° while drift rises by 2 pp. Enabling only the accelerometer kernel gives a similar orientation error of 7.6°, so it accounts for most of the orientation improvement (Section IX-C). For the aliased returns, full weighting increases fast-racing errors and deletion costs fast-racing drift, so the measured down-weight is kept.

Priors. Replacing the measured bias priors with hand-set values increases slow-racing velocity error by 5% and backflip orientation error by 1.4°, but reduces fast-racing velocity error by 4%. We retain the measurement-based priors because their widths reflect observed bias variability. The position-row staleness has negligible effect on racing and reduces the backflip orientation error by 1.2°. Re-applying the window-start priors at every stride cuts the fast-racing drift from 1.33 to 0.92% by holding the vertical velocity in place, but doubles it on slow racing and on the flip flight and double-counts the previous estimate (Section VI-F).

Regularization. Removing the smoothness priors substantially improves backflip velocity and position accuracy, with a smaller improvement in orientation (velocity error 1.66 to 0.76 m/s, orientation 12.1 to 10.5°, drift 2.9 to 1.1%), but increases fast-racing position drift to 5.7%, predominantly in the vertical channel. The shared configuration therefore retains them (Section IX-F).

IX. LIMITATIONS

A. Evaluation Scope

Each flight regime is evaluated on a *single* representative recording, so the headline numbers carry no between-flight error bars. The held-out flights (Section VIII-B) and the cross-platform validation (Section VIII-C) broaden the evidence, but they do not substitute for repeated trials.

Attributable AI wording: Table 5.1 — Opus 5

TABLE VI
FRONT-END AND PRIOR ABLATION (LIVE EDGE: VELOCITY / ORIENTATION / DRIFT)

Configuration	Slow racing	Fast racing	Backflips
Deployed	**0.16 / 1.65 / 0.32**	**0.28 / 2.56 / 1.33**	**1.66 / 12.1 / 2.9**
No prefilter	0.20 / 1.66 / 0.88	0.42 / 2.63 / 1.70	1.61 / 12.1 / 2.4
Robust kernels on (δ=1, δ_a=2)	0.16 / 1.65 / 0.32	0.29 / 2.69 / 1.13	1.96 / 7.38 / 4.9
Aliased kept (w=1)	0.17 / 1.67 / 0.27	0.37 / 2.86 / 2.17	1.61 / 12.1 / 2.9
Aliased deleted	0.16 / 1.65 / 0.35	0.29 / 2.55 / 1.65	1.66 / 12.0 / 3.1
Bias priors hand-set (10^4)	0.17 / 1.65 / 0.36	0.27 / 2.58 / 1.25	1.68 / 13.5 / 2.9
Position staleness off	0.16 / 1.64 / 0.32	0.28 / 2.50 / 1.35	1.74 / 13.3 / 2.8
Window-start priors every stride	0.16 / 1.69 / 0.60	0.28 / 2.41 / 0.92	1.69 / 12.5 / 5.2
Smoothness priors off	0.16 / 1.67 / 0.34	0.23 / 2.52 / 5.72	0.76 / 10.5 / 1.1

The evaluation covers indoor flights on two platforms using the same radar chip. The main laboratory recordings last 18–26 s. Long-duration operation, different radar hardware, and performance in smoke or fog remain untested. The public flights use VINS pseudo ground truth, and the backflip recording contains MoCap tracking artifacts (Section VII).

B. Heading Sensor Requirement

Yaw is unobservable from Doppler+IMU (Section VI-B). Here, reference yaw serves as an idealized magnetometer surrogate. The evaluation therefore does not establish performance with a real heading sensor, including its bias and time-varying magnetic disturbances. The weight is set at what a real sensor would justify: a flight magnetometer resolves heading to a few degrees [25] with strongly time-correlated errors, which under the correlation-time protocol of Section V-G gives a weight of order 1–10; we deploy $\lambda_\psi = 0.6$, the conservative end of that range. With $\lambda_\psi = 0$ (no heading information after the start gauge) yaw drifts within the flight: the orientation error grows to 10 and 9° on the two racing flights, with 0.5–0.6 m/s of world-frame velocity error. A ten times stronger weight reduces orientation error by 0.03–0.2° on racing and 2.8° on the flip flight, but increases drift by up to 0.2 pp. We retain the weaker weight to limit reliance on reference heading, accepting the orientation trade-off.

C. Extreme Dynamics: Backflips

The backflip flight circles in the horizontal plane while looping continuously in the vertical plane at body rates around 10 rad/s. Section V characterizes the radar errors in this regime: 38% of its returns alias, the noise floor of the per-return law is four times the deployed nominal value, the frame-shared error and the off-boresight excess the law does not carry are at their largest. One possible contributor is intra-frame motion. During a flip the scene sweeps some 17° across a frame's integration time, and without per-return timestamps the returns of one frame cannot be placed on the trajectory individually.

The configuration also contributes to the backflip errors. Robust accelerometer weighting improves orientation at the cost of drift, while removing smoothness priors improves backflip accuracy but worsens fast-racing drift (Section VIII-E). We retain the shared configuration for its racing performance and report the backflip degradation under the same settings.

D. Sustained Flight Beyond the Doppler Ambiguity

Every aliased return has to be assigned a wrap. This is based on the last radar frame's consensus velocity, integrated forward with the IMU (Section VI-C). While only a few returns are aliased the consensus rests on the unambiguous majority, but when most of the frame is aliased it is built from the estimated unwraps themselves, and an error can enter the consensus. This then displaces the next frame's prediction by one wrap, $2v_{\max} = 6.3$ m/s, and enters the cost quadratically, so down-weighting does not help and a frame agreeing on the same wrong wrap still passes the prefilter. The held-out circular flight at 5 m/s is this regime (Section VIII-B).

E. Vertical Channel

On fast racing the vertical channel carries most of the drift (vertical RMSE 0.51 m against 0.27 m horizontally, Fig. 6). On slow racing the corresponding errors are 0.09 and 0.11 m. Sweeping the locked radar pitch from 25.5 to 29.5° moves the vertical drift monotonically in opposite directions on the two racing flights. This sweep does not identify a common correction, so we keep the measured pitch. The prefiltered racing returns also show no constant elevation bias (Section V-B).

The fast-racing vertical error remains 0.51 m with or without position-prior inflation. The source of the remaining vertical drift is unresolved.

F. Fixed Parameters and Motion Priors

Two constants of the radar noise model are flight properties held fixed at nominal-regime values: the noise floor σ_0 of the per-return law (measured 0.049–0.327 m/s across regimes, deployed at the racing geometric mean) and the frame-shared term σ_c (measured 0.83–1.55, deployed at 1.27). Both, like every constant of the model, are properties of this radar and this IMU: a different sensor requires the same measurements, and Section V is that procedure. Online estimation of these terms is a possible extension, not implemented here: σ_0 is a within-frame statistic and can be measured in flight without a reference, whereas σ_c is a frame-mean statistic whose in-flight estimate would have to separate the shared measurement error from trajectory error.

The fixed motion priors, spline grids and window length impose a trade-off between racing and backflip performance (Section VIII-E) and also constrain the accelerometer bias estimates. A motion prior derived from measured motion statistics is one possible alternative to the fixed smoothness penalties, but remains untested.

G. Marginalization and Covariance

The marginalization prior is re-centered at each warm start with its gradient set to zero, retaining only its curvature. This does not preserve the full Gaussian marginal. Among factors touching leaving states, those extending beyond the retained boundary are dropped. Angular-travel inflation approximates the effect of stale linearizations (Section VI-F). The NEES results show conservative covariances on racing, but do not establish consistency across regimes (Section VIII-D).

Attributable AI wording: Fable 5.1 Opus 5

H. Knot Density

Finer orientation knots do not improve the tested backflip estimates. At the deployed 16 ms the orientation spline represents the MoCap attitude through the flips to 1.3°, the same at 8 and 4 ms. In the coarser-grid sweep, backflip orientation error is 11.9–12.8° from 8 to 64 ms and increases to 14.3° at 128 ms, with λ_α held fixed across the sweep. These results suggest that the orientation grid could be coarsened where compute is tight.

X. Conclusion

Can a single-chip mmWave radar plus IMU provide usable 6-DoF state estimation? Yes, as a *velocity and orientation* source with heading aiding: at the live leading edge, 0.16 m/s / 1.7° on gentle racing and 0.28 m/s / 2.6° on aggressive racing, with position drifting at 0.3–1.3% of distance. The same configuration is used for all evaluated flights and platforms. The solver runs faster than its 0.3 s stride on a laptop CPU, transfers unchanged to a held-out flight and to a public dataset, and reports a covariance that is conservative on racing (Section VIII-D). Absolute position is not observable from Doppler+IMU and drifts with distance, and yaw requires a heading sensor. Through sustained flips the estimate degrades, reflecting both radar measurement limitations and the racing-oriented configuration.

Within those structural limits, the answer rested on three things. First, *characterizing the sensor before designing the estimator*: the radar's errors have a Gaussian core whose width is a per-return quantity. Doppler error from bearing uncertainty grows linearly with speed, couples through the transverse velocity and increases off boresight, with a fitted constant of about one angle-FFT bin. The errors also include a component shared by all returns of a frame, an outlier remainder, and aliased returns with elevated bearing-related residuals even after Doppler unwrapping. Second, *deriving the radar noise weights and the radar and IMU stream weights from those measurements*, and the bias priors and the staleness of the marginalization prior from measurements of their own, so that the reported covariance is a prediction tested against ground truth (Section VIII-D). Third, *a fixed-lag smoother* that carries past information through an approximate marginalization prior, built from exactly the factors that leave the window, applied without a global scale on every regime. The largest gains came from measuring and re-weighting the existing measurements.

Future directions follow from the limitations. More precise per-return radar timing could help model intra-frame motion during backflips. The noise floor σ_0 and the frame-shared term σ_c could be adapted in flight, but estimating the shared component requires separating measurement error from trajectory error. Repeated recordings per regime would quantify between-flight variability. An absolute-height anchor such as a floor-plane factor [12] could be gated on the frames that observe the intermittently visible floor.

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